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AI-based traffic control and management systems - a synthesis of Macedonian research experiences

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Abstract

Under the influence of advances in Artificial Intelligence and Machine Learning technology, the transport sector is on the brink of a major revolution. Algorithms from these areas have been implemented in various fields, from autonomous vehicles, and innovative tools for gathering and visualizing traffic data for in-depth analysis of traffic flow characteristics, to the next generation of intelligent traffic control and management systems. Through two case studies, this paper aims to present the synthesis of Macedonian research experience gained over a long process of exploring the opportunities and challenges of development and integrating emerging technologies into the transport system. The first case study describes the implementation of a reinforcement learning (RL) algorithm in the design and testing of an adaptive control strategy at urban intersections. The second case study involves the application of a sophisticated AI-based tool for traffic flow data gathering, by video data processing. In both cases, the process of testing and evaluation is obtained in the simulation environment. The obtained simulation results demonstrated the efficiency of the implemented solutions in decreasing traffic congestion, and vehicle emissions, and increasing throughputs which benefits society as a whole.

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1. Introduction

There are many implementations of AI in various fields of the transport sector and transport services due to their capability to act on time to every circumstance's change is a real challenge for the transport system. Artificial intelligence (AI) techniques can create a robust control environment through faster response and through increased flexibility in adapting to changes in the system [Adler and Blue, 2002]. And just as the power of technology is expected to continue future operational improvements will be continued. One of the most popular applications of AI is the development of autonomous vehicles. The combination of AI algorithms, advanced sensor technologies, and deep learning techniques has proven the potential to revolutionize traffic allowing better control of vehicles and navigation without driver assistance. By the analysis of vast amounts of real-time data from multiple sources (area-wide surveillance systems, detector systems, GPS systems, etc), AI will continue to progress in the area of traffic flow forecasting and management. This will enable improved effectiveness of the traffic systems via dynamic routing, prevention of traffic congestion, and adaptive traffic signal control. A very popular technique among researchers, intelligent agent technology is an effective tool due to its capability to describe at an individual level every autonomous entity on complex networks. Some of the most popular applications are in the traffic control centres where they can successfully manage the traffic lights to achieve the best signal plan for a road network.

New levels and stages of intelligent infrastructure systems are created by AI implementation. Namely, communication vehicle-to-infrastructure (V2I) and vehicle-to-vehicle (V2V), can contribute to greater road safety, efficiency, and coordination. Different types of customized transport solutions (multimodal trip planning, on-demand ride-sharing) and personalized mobility services are provided by the use of AI algorithms.

Generally speaking, in the development of sustainable transport systems, by utilizing ML algorithms energy consumption can be optimized, by promoting environmentally friendly driving behaviours and contributing to the integration of new generations of autonomous vehicles on the roads. Routing systems will also prioritize eco-friendly options to reduce emissions and promote sustainable mobility.

Safety on roads is a prime concern. AI will continue to advance traffic safety systems by identifying potential vehicle and infrastructure problems and minimizing the risk of vehicle accidents by enabling proactive maintenance.

In the field of traffic data collection and surveillance data, in-depth analysis, and simulations by combining AI and big data analytics today we can create more convenient simulation models and new solutions as a result of the detailed and accurate data, for example, the Good Vision tool (GoodVision Ltd, 2023).

The authors' research activities related to AI-based traffic solutions are long-lasting and in continuation of the paper, we will provide the synthesis of Macedonian research experience in the development, testing, and integration of AI techniques and technologies into transport systems. A case study methodology is applied; field data and simulation tools are used in both cases for operational analysis of the tested solutions. The presented synthesis aims to explore the opportunities, challenges, and benefits regarding the implementation of the newest techniques and technologies to cope with increased traffic problems. Furthermore, to overcome the limited vision of the old traditional approaches in dealing with traffic problems which are still dominant in Macedonian and Western Balkan countries in general, and to recommend investing in Intelligent Transport Systems and innovative technologies [Koltovska Nechoska et al., 2023.]. The first use case is related to the implementation of an RL algorithm in the design and testing of adaptive control at urban intersection. The second case study involves the application of a sophisticated AI-based tool for traffic flow data gathering, by video data processing. The process of testing is realized in the simulation environment. Except for delay, throughput, level of service, etc., the environmental effect was also evaluated for both case studies by using the EnViVeR traffic emission simulator. The paper ends with a description of the conclusion and future work.

2. Control of Urban Intersection by Q-learning based Strategy

This section presents the implementation of the reinforcement learning (RL) algorithm in the design and testing of an adaptive control strategy at urban intersection.

2.1. Applied method

Intelligent agent technology is a very promising available technique and well-suited for real-time applications as a result capability of fast responses to various traffic states. One of the most popular techniques is Reinforcement Learning (RL) or learning from experience.

RL does not require supervision and it is goal-directed learning; from interaction with an environment and how to map situations to actions to maximize a numerical reward (Sutton and Barto, 1998).

RL has a broad range of algorithms that use the same principle; in the research, we used the Q-learning algorithm (control algorithm of the temporary difference (off-policy TD Control)) [Sutton and Barto, 1998; Abdulhai et.al, 2003].

The process of interaction between agent and environment is as follows:

1. The agent obtains a reward $r(s,a)$ for taken action a in state s , depending on how this activity affects the environment.
2. The previously calculated Q -value (as of time $n-1$) is updated recursively using state s , action a , and reward $r(s,a)$:

$$\hat{Q}_n(s,a) \leftarrow (1 - \alpha_n) \hat{Q}_{n-1}(s,a) + \alpha_n [r + \max_{a'} \hat{Q}_{n-1}(s',a')]$$

where $\hat{Q}_{n-1}$ is the prior estimate of the Q -value of taking action a for the state s , and $\max(\hat{Q}_{n-1}(s',a'))$ is the prior estimated Q -value of the optimum policy that is starting in the state s' .

The adaptive control strategy is performed by an agent; our “agent” is a software-based computer program named VSR -VISSIM Simulation Runner, which supports an autonomous decision-making process (Fig.1). The user interface consists of three components.

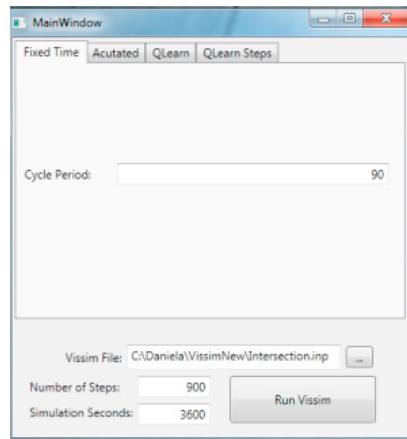


Fig.1. User interface of the VSR tool.

The first one is related to the type of Signal control. Namely, we can choose three types of control: Fixed-time, Actuated, and Q-learning-based control. In this paper, the evaluation of the effectiveness of Q-learning based control is compared with fixed time control.

The second component is related to the signal Cycle Period; different values of the signal cycle period can be set in the chosen field. The cycle length in our case is 90 s.

The third component is related to the connection with the VISSIM file, the number of iterations, and the simulation parameters. The total number of iterations for agent training is 900 and the simulation period of testing is 3600s. After initializing the VISSIM file (Run Vissim) the simulation process starts.

2.2. Evaluation and testing

The Q-learning-based signal control strategy was implemented on the four-leg intersection located in the central area of Bitola. The intersection was created using the traffic microscopic simulator VISSIM as illustrated in Fig 2.



Fig. 2. Analysed intersection.

The traffic data were collected manually for the afternoon peak hour (15-16h). The total flows for the examined intersection are presented in Table 1. The total traffic demand is 2 273 vehicles/h.

Table 1. Traffic flow.

	Intersection approach											
	I			II			III			IV		
	Left	Through	Right	Left	Through	Right	Left	Through	Right	Left	Through	Right
Total demand/lane	68	610	32	61	162	248	138	472	228	37	130	87
Total demand/approach	710			471			838			254		
Total demand (vehicle/h)	2, 273											

Detectors are located on all lanes. The simulation was conducted for a peak time period. All traffic parameters: occupancy, gap, and vehicle count, are measured directly by the simulator. Traffic demand at the links was created with Poisson distribution.

The RL agent observes only one state, performs one action and receives a reward. States are defined as gaps, phases, and occupancy detected at the intersection approaches. Actions are defined as extensions of the green phase or change of the signal phase; total throughput is the reward function. The selection of the action is based of ϵ -greedy policy and the value 0.01. was chosen for epsilon in this case (Ivanjko & Koltovska, 2006).

To evaluate the effectiveness of the Q-learning based strategy the following MoEs are defined: total average delay, total number of stops, and total throughput (Table 2).

The Q-learning based strategy testing has been performed in comparison with fixed-time control under the following scenarios.

Scenario 1: Uncongested traffic conditions and unknown traffic demand;

Scenario 2: Congested traffic conditions and unknown traffic demand.

Table 2. Comparative analysis.

Measure of effectiveness	Scenario 1		Scenario 2	
	Fixed-time control	Q-learning based control	Fixed time control	Q-learning based control
Total average delay (s)	18,485	11,683	43,211	31,795
Total number of stops	1,667	1,230	3,005	2,513
Total throughput	2,310	2,549	2,907	3,308

The analysis of results shows that Q-learning based signal control is dominant versus fixed time control.

The vehicle emission simulator EnViVeR was used to compute the emission values regarding CO₂ and NO_x gasses and (PM₁₀) - particles with 10 µm diameter (Table 3).

Table 3. Comparative analysis.

Emission type	Scenario 1		Scenario 2	
	Fixed-time control	Q-learning based control	Fixed-time control	Q-learning based control
CO ₂ [g/km]	213.5	156.7	317.8	278.6
NO _x [g/km]	0.5254	0.3321	0.8144	0.6111
PM ₁₀ [g/km]	0.03445	0.02214	0.06229	0.04451

As we expected, emissions values are lower under Q-learning and according as a result of the capability of adaptation to traffic flow conditions in real-time.

3. Traffic Data Collection, Analysis and Simulation by AI-based tool

Obtaining accurate and detailed information about traffic conditions is of particular importance for traffic engineers and researchers for effective analysis in traffic management and control, planning, and design. Conventional methods for collecting traffic data mostly are performed manually, which takes a lot of time and requires significant use of financial and human resources.

In Macedonia, the manual collecting method is still dominant. With the constant development of AI-based technologies, especially in recent years, various applications have been also developed for traffic data collection and surveillance that are valuable inputs into the design development, and testing of traffic control and management systems and strategies into simulation environments, such as one of the newest application - Good Vision provided by company GoodVision Ltd, Fig.3 [Koltovska et al, 2023].

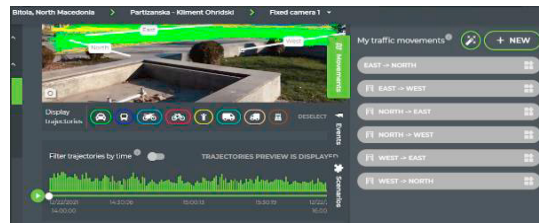


Fig. 3. Good Vision – interactive dashboard.

3.1. Evaluation and testing

The data collection process was realized for the intersection shown in Fig. 4 during the peak hour period from 14-16h [1] and the VISSIM model was created to obtain intersection performances as well as environmental impact estimation.



Fig. 4. Analysed intersection.

The obtained video recordings from the field were processed in Good Vision's Video Insights platform which resulted in acquiring detailed traffic data regarding the total number of vehicles, entry/exit time at the intersections, and travel time Fig. 5 and Fig. 6.

APPROACH	East (Approach II)				North (Approach I)				South (Approach III)				West (Approach IV)			
MOVEMENT	East -> North	East -> South	East -> West	Approach total	North -> East	North -> South	North -> West	Approach total	South -> East	South -> North	South -> West	Approach total	West -> East	West -> North	West -> South	Approach total
12/22/2022 14:00:00 - 12/22/2022 14:15:00	3	6	48	57	1	19	65	85	8	45	57	110	99	11	57	167
12/22/2022 14:15:00 - 12/22/2022 14:30:00	1	2	57	60	3	16	54	73	11	20	68	99	115	22	67	204
12/22/2022 14:30:00 - 12/22/2022 14:45:00	1	2	45	48	3	12	50	65	9	15	72	96	101	15	61	177
12/22/2022 14:45:00 - 12/22/2022 15:00:00	3	3	63	69	3	14	58	75	3	18	64	85	115	15	54	184
12/22/2022 15:00:00 - 12/22/2022 15:15:00	2	3	55	60	3	16	56	75	11	17	62	90	119	14	56	189
12/22/2022 15:15:00 - 12/22/2022 15:30:00	2	4	75	81	4	12	56	72	8	30	43	81	113	15	69	197
12/22/2022 15:30:00 - 12/22/2022 15:45:00	7	4	62	73	2	15	64	81	7	19	51	77	100	13	53	166
12/22/2022 15:45:00 - 12/22/2022 16:00:00	1	2	22	25	1	11	20	32	3	6	17	26	48	12	25	85
12/22/2022 16:00:00 - 12/22/2022 16:15:00	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
SUM	20	26	427	473	20	115	423	558	60	170	434	664	810	117	442	1369
MOVEMENT	East -> North	East -> South	East -> West	Approach total	North -> East	North -> South	North -> West	Approach total	South -> East	South -> North	South -> West	Approach total	West -> East	West -> North	West -> South	Approach total
APPROACH	East				North				South				West			

Fig. 5. Total vehicle counts represented by using the GoodVision.

CLASS	TIME GAP (s)	ENTRY TIME (East)	EXIT TIME (South)	TRAVEL TIME (s)
CAR	6.52	2022-12-22T14:11:37.660	2022-12-22T14:12:07.620	29.96
CAR	51.72	2022-12-22T14:12:29.380	2022-12-22T14:12:35.460	6.08
CAR	6.24	2022-12-22T14:12:35.620	2022-12-22T14:12:56.758	21.139
CAR	16.56	2022-12-22T14:12:52.180	2022-12-22T14:13:04.773	12.594
CAR	4.08	2022-12-22T14:12:56.260	2022-12-22T14:13:02.706	6.447
TRUCK	2.96	2022-12-22T14:12:59.220	2022-12-22T14:13:07.180	7.96
CAR	55.08	2022-12-22T14:13:54.300	2022-12-22T14:14:10.380	16.08
CAR	11.12	2022-12-22T14:14:05.420	2022-12-22T14:14:11.980	6.56
CAR	3.32	2022-12-22T14:14:08.740	2022-12-22T14:14:18.820	10.08
VAN	35.32	2022-12-22T14:14:44.060	2022-12-22T14:15:15.900	31.84
CAR	10.32	2022-12-22T14:14:54.380	2022-12-22T14:14:57.158	2.779
CAR	17.56	2022-12-22T14:15:11.940	2022-12-22T14:15:18.420	6.48
CAR	5.88	2022-12-22T14:15:17.820	2022-12-22T14:15:22.580	4.76
CAR	9.6	2022-12-22T14:15:27.420	2022-12-22T14:15:31.180	3.76
CAR	43.28	2022-12-22T14:16:10.700	2022-12-22T14:16:56.997	46.298
VAN	2.56	2022-12-22T14:16:13.260	2022-12-22T14:16:15.500	2.24
CAR	24.84	2022-12-22T14:16:38.100	2022-12-22T14:16:59.340	21.24
CAR	27.561	2022-12-22T14:17:05.661	2022-12-22T14:18:00.820	55.159
CAR	6.319	2022-12-22T14:17:11.980	2022-12-22T14:17:13.980	2
CAR	39.72	2022-12-22T14:17:51.700	2022-12-22T14:18:03.552	11.852

Fig. 6. Total number of vehicles, entry/exit time, and travel time represented by using Good Vision.

We need to emphasize that due to the unsuitable angle of the video cameras, some of the data presented in Fig. 6 are incomplete. The authors recommend that the placement of the camera capturing the traffic should be higher and unobstructed to achieve continuous and uninterrupted processing of the recordings.

Obtained results of average queue length, maximum queue length, level of service, and average delay/vehicles are presented in Table 4.

Table 4. Obtained measure of effectiveness.

Measure of effectiveness (MoE)	Obtained Values
Average queue length (m)	7.51
Level of service (LOS)	LOS B
Average delay/vehicle (s)	19.80

From the above-presented results, we can conclude that the intersection is operating under a high level of service B which had have positive impact on the environment (Table 5).

Table 5. Obtained vehicle emissions.

Emission type	Obtained values
CO ₂ [g/km]	128.5
NO _x [g/km]	0.3753
PM ₁₀ [g/km]	0.02146

4. Conclusion

The development and implementation of AI in the transport sector is a complex and often controversial process. The pace of AI development is likely to be linked to factors such as (1) widespread recognition of the need to reduce congestion and 2) advances in the spread-spectrum of technology.

AI is transforming transport by providing cutting-edge solutions that improve safety, efficiency, and sustainability.

Through the implementation of two case studies opportunities, advantages, and constraints of implementing new technology to address traffic issues are presented. The article is meant to provide a proof of concept and/or feasibility evaluation of applied methods and the future research steps will be continued toward the integration of Q learning-based control strategy and Good Vision data as e key ingredients to creating a powerful intelligent traffic management and control system.

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